

Course: B. Tech.

Branch : All Branches

Semester : I

Subject Code & Name: Computer Programming in C [BTES104]

Max Marks: 60

Date: 17/08/2023

Duration: 03:00 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Attempt the following questions.

[12]

- A) Define the flowchart. Explain the use of any Five flowchart symbols with the diagrammatic notation. Understand
- B) What do you mean by the algorithm? Write a simple algorithm to add two integers. Convert the algorithm written for addition of two numbers into the flowchart. Application

Q.2 Solve any TWO Questions.

[12]

- A) Explain the following operators used in the programming: Understand
a) Bitwise Operator b) Relational Operator c) Arithmetic Operator
- B) Explain the Conditional Operator with proper syntax. Also, write a program in C to check the input given by the user is either even or odd using the conditional operator. Application
- C) What is the purpose of precedence and associativity in the programming? Write the precedence wise along with the associativity for the following operators: ^ , &&, ||, ? :, = +, ++. Understand

Q. 3 Solve any TWO Questions.

[12]

- A) Explain the while and do.....while loop with writing a program in C program to print the sum of first 5 natural numbers. Understand
- B) Define a Function. Enlist and explain the types of the function. Write a Syntax of function definition, function call, and function prototype in C. Remember
- C) Write a program in C. Two integer numbers are given as a input to program and the output of the program is addition of the two integers using function which returning the addition of the given numbers. Application

Q.4 Attempt the following questions.

[12]

- A) Define the array. Enlist and explain the types of array. Write the syntax for following for the one dimensional array: Remember

- a) declaration of array
- b) initialization of array
- c) declaration with initialization

B) Write a program in C to find the average of n numbers given as an input by user **Application** using array. Also, display on the average of n numbers on the screen.

Q. 5 Attempt the following questions.

[12]

- A)** Explain the structure with proper syntax. With proper syntax give an example for **Remember** declaring the structure variable. Write the syntax for nested structure with proper example.
- B)** Write a C program to Store Information of Students and display the information **Application** on screen using structure.

******* END OF PAPER *******